

Week 6 - Factorial Experiments

Department of Statistical Sciences, University of Cape Town

Factorial Experiments

- So far: single treatment factor
- Variation in response cannot be explained by a single factor at a time
- Especially when factors interact

An interaction occurs when the effect of one factor depends on the level of the other factor.

Factorial experiments

- More than one treatment factor
- Each treatment factor has levels
- Factor A has a levels and Factor C has c levels
- Complete factorial experiment: $a \times c$ treatments

Example of factorial studies

- The effect of playback speed and lecture modality on comprehension of lectures.
- The effect of advertising medium (Email list, instagram and television ads) and discount level (10% and 20%) on product sales.
- The effect of fertiliser type (F1,F2) and irrigation level (I1,I2) on wheat yield.

Factorial treatment structure and designs

- Can be either a CRD or RCBD
- Fertiliser: F1, F2
- Irrigation: I1, I2
- Treatments: F1I1, F1I2, F2I1, F2I2
- 2×2 Factorial experiment

Completely Randomised Design

A 4x4 grid representing a completely randomised design. The grid is outlined in green and contains handwritten labels in various colors (orange, red, blue, black) for each cell. The labels represent combinations of two factors, F_1 and F_2 , each with two levels, I_1 and I_2 . The labels are distributed as follows:

$F_1 I_2$	$F_2 I_1$	$F_2 I_1$	$F_2 I_2$
$F_2 I_1$	$F_1 I_1$	$F_1 I_2$	$F_1 I_1$
$F_2 I_2$	$F_1 I_1$	$F_2 I_2$	$F_1 I_2$
$F_2 I_1$	$F_2 I_2$	$F_1 I_2$	$F_1 I_2$

Randomised Block Design

$F_1 I_1$	$F_2 I_2$	$F_1 I_2$	$F_1 I_2$
$F_2 I_2$	$F_2 I_1$	$F_1 I_1$	$F_2 I_1$
$F_2 I_2$	$F_2 I_2$	$F_1 I_1$	$F_1 I_2$
$F_2 I_1$	$F_1 I_1$	$F_1 I_2$	$F_2 I_1$

Replication

- Notice each treatment appeared more than once
- RBD: each treatment appeared twice in a block, so four times in total
- CRD: each treatment appeared four times
- Replication of treatments is NB in factorial experiments (more than two experimental units)
- Allows us to estimate the interaction effect

Interactions

- Effect of a factor depends on the level of another factor, they interact
- Define effect
- Regression: effect of continuous EV = slope.
- Regression: effect of categorical EV = change in response relative to baseline.
- ANOVA model: change relative to overall mean.

Main effects

- A main effect is the effect of a factor (e.g. Fertilizer) on the response.
- Numerically, we estimate it for each level of the factor, as the difference between that level's mean (averaged over the other factors) and the overall mean.

Broken down:

1. average change in response,
2. averaged over all levels of the other factors
3. relative to the overall mean.

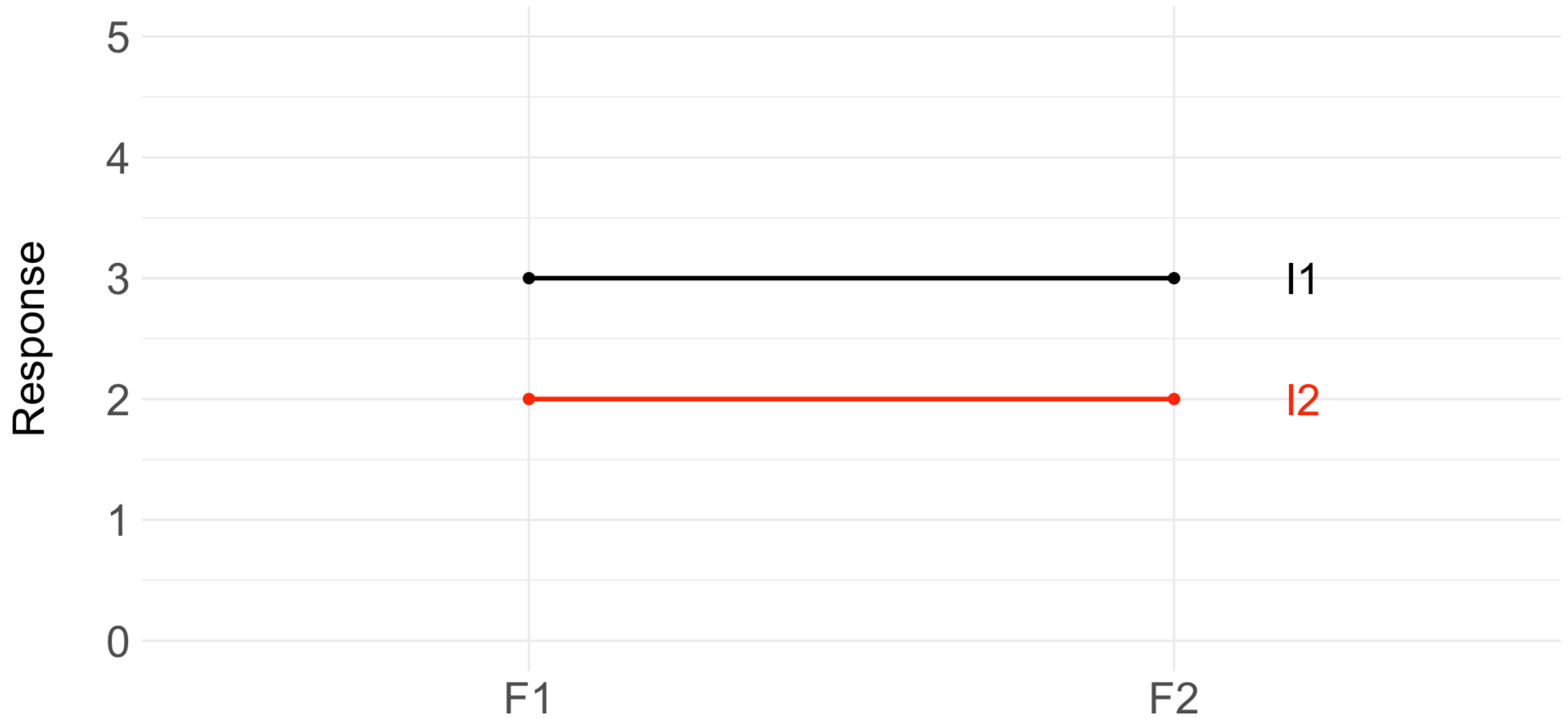
Interaction effects

- A main effect is tied to a single factor;
- an interaction effect is tied to a combination of factors (i.e. treatment).

Interaction effect:

1. Change in response relative to the main effects.

Interaction plots



How to identify main effects

Main effects:

1. Are there main effects of Fertiliser? Look at the average change in response averaged over the levels of irrigation.
2. Are there main effects of Irrigation? Look at the average change in response averaged over the levels of fertiliser

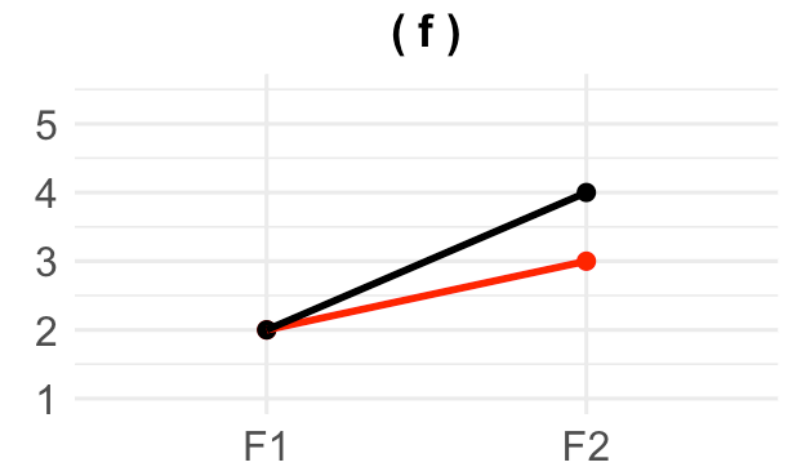
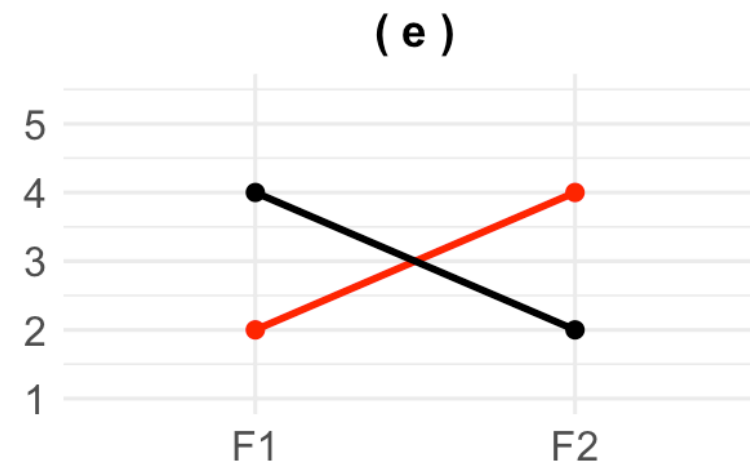
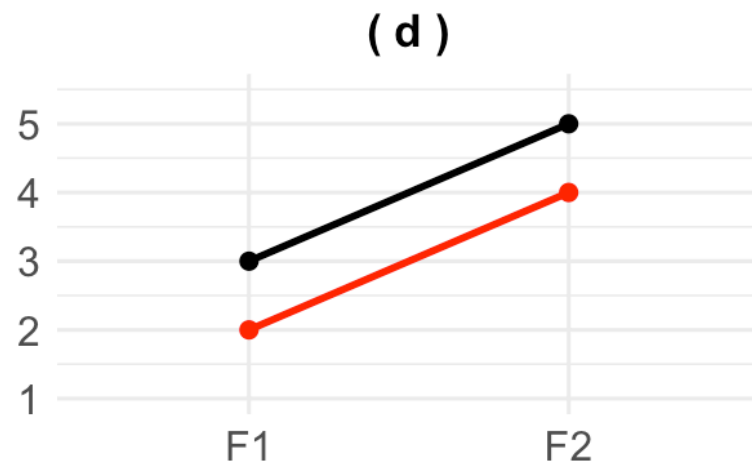
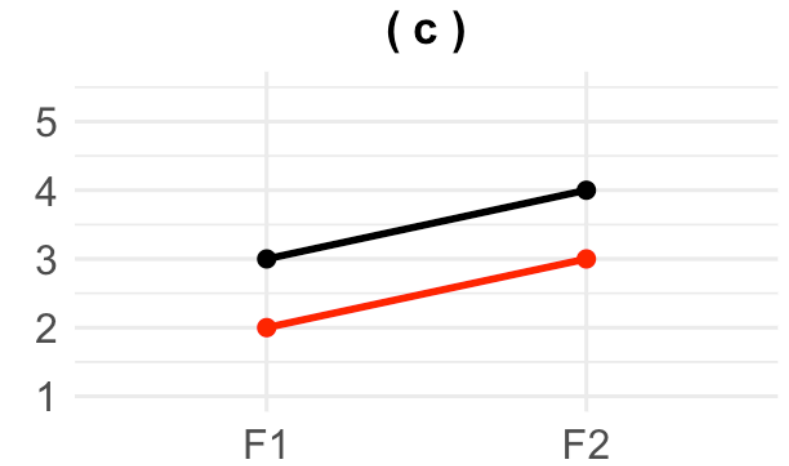
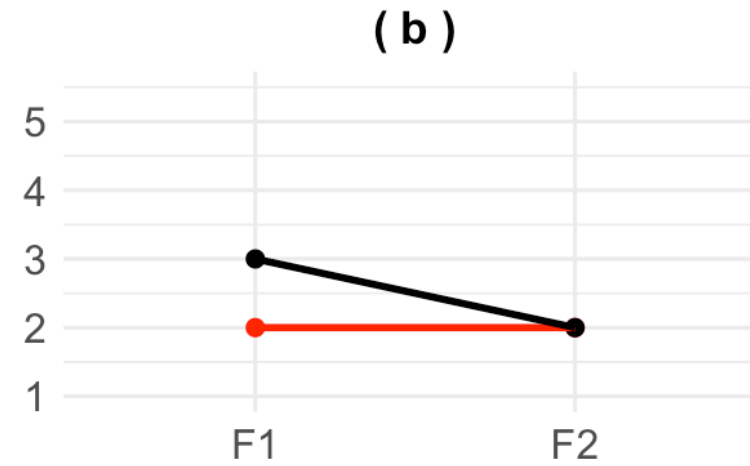
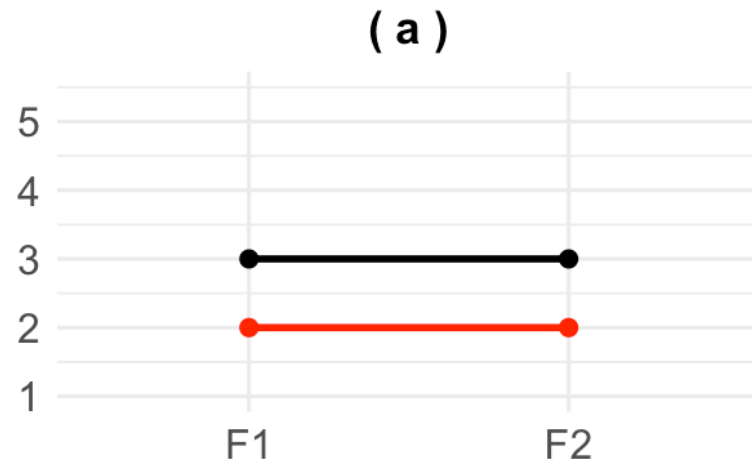
How to identify interaction effects

Interaction effects:

Does the effect of fertiliser on the response depend on the irrigation level? Or does the effect of irrigation level on the response depend on the fertiliser type?

- Does changing from F1 to F2 differ at each level of irrigation?
- Does changing from I1 to I2 differ at each level of fertiliser?

More plots!



Factor B —●— I1 —●— I2

Interaction plots

- Qualitative sense of effects: are there main effects present? Is there an interaction?
- Assess this with plots by looking at the relative differences between treatment means
- Doesn't give us the numerical size of effect which is calculated relative to the overall mean.

Model for factorial experiments

- Two treatment factors (A and C) and no blocking factors:

$$Y_{ijk} = \mu + A_i + C_j + (AC)_{ij} + e_{ijk}$$

- Y_{ijk} is the k^{th} observation of the $(ij)^{th}$ treatment.
- $(AC)_{ij}$ is the interaction between the i^{th} level of A and the j^{th} level of C (single symbol and not the product of main effects)
- Assumptions depend on the design used.

Example

- We are studying the effects of **Drug** (Factor A) and **Diet** (Factor B) on blood pressure reduction.
- Factor A: Drug (Yes, No)
- Factor B: Diet (Healthy, Unhealthy)
- Treatments?

Data

Responses for Each Treatment
Combination

Diet	No Drug	Yes Drug
Healthy	1.5	7.8
	2.2	8.1
	2.3	8.1
Unhealthy	3.8	5.8
	4.1	6.0
	4.2	6.2

Means

Treatment Means for 2×2 Factorial
Design

Drug	Diet	MeanResponse
Yes	Healthy	8
Yes	Unhealthy	6
No	Healthy	2
No	Unhealthy	4

Main effects of Factor A

- Overall mean: $\frac{2+4+8+6}{4} = 5$
- Main effects of factor A Drug

Level 1:

Mean of treatments with Drug = $\frac{8+6}{2} = 7$

$$\hat{A}_1 = 7 - \mu^{\wedge} = 7 - 5 = 2$$

Level 2:

Mean of treatments without Drug = $\frac{2+4}{2} = 3$

$$\hat{A}_2 = 3 - \mu^{\wedge} = 3 - 5 = -2$$

Main effects of Factor C

- Overall mean: $\frac{2+4+8+6}{4} = 5$
- Main effects of factor C Diet

Level 1:

Mean of treatments with Healthy diet = $\frac{8+2}{2} = 5$

$$C_1^{\wedge} = 7 - \mu^{\wedge} = 5 - 5 = 0$$

Level 2:

Mean of treatments without Unhealthy diet = $\frac{4+6}{2} = 5$

$$C_2^{\wedge} = 3 - \mu^{\wedge} = 5 - 5 = 0$$

The interaction effect

- Do the observed treatment means equal what we would expect based on just main effects?
- In other words, as if the factors don't interact?

Expected mean of Treatment 11 (Yes and Healthy): $\mu^{\wedge} + A_1 + C_1 = 5 + 2 + 0 = 7$

Observed treatment mean: $\bar{Y}_{11} = 8$

Extra bit: $\bar{Y}_{11} - (\mu^{\wedge} + A_1 + C_1) = 8 - 7 = 1$

- So the observed treatment mean is 1 unit higher than what we would expect if the factors worked independently (i.e., additively).
- This extra effect is the interaction — it captures how the combination of Drug and Diet together changes the response beyond what their individual effects explain.

The interaction term

- The interaction effect is calculated as:

$$(AC)_{ij} = \bar{Y}_{ij.} - (\mu + A_i + C_j)$$

- If we don't have replication of treatments, then we cannot estimate this term.
- Can show with one observation per treatment, the error term is the same as the interaction term.
- Need replication to separate them!

Decomposition of sums of squares

$$Y_{ijk} - \mu = A_i + C_j + (AC)_{ij} + e_{ijk}$$

Now, each of the terms on the RHS is a deviation from a mean.

- The main effects from the overall mean μ ,
- The interaction effects from $\mu + A_i + C_j$ (expected mean if no interaction effects),
- The error terms from the treatment means.

$$SS_{total} = SS_A + SS_C + SS_{AC} + SS_E$$

ANOVA Table

Source	SS	df	MS	F
A Main Effects	$SS_A = nc \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2$	$(a - 1)$	MS_A	$\frac{MS_A}{MS_E}$
C Main Effects	$SS_C = na \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2$	$(c - 1)$	MS_C	$\frac{MS_C}{MS_E}$
AC Interactions	$SS_{AC} = n \sum_{ij} (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$	$(a - 1)(c - 1)$	MS_{AC}	$\frac{MS_{AC}}{MS_E}$
Error	$SS_E = \sum_{ijk} (Y_{ijk} - \bar{Y}_{ij.})^2$	$ac(n - 1)$	MSE	-
Total	$SS_{total} = \sum_{ijk} (Y_{ijk} - \bar{Y}_{...})^2$	$abn - 1$	-	-

Hypotheses

There are three F-tests in this ANOVA table.

1. $H_{AC} : (AC)_{ij} = 0$ for all i and j (Factors A and B do not interact)
2. $H_A : A_i = 0 \quad i = 1, \dots, a$ (Factor A has no main effects)
3. $H_B : B_j = 0 \quad j = 1, \dots, b$ (Factor B has no main effects)

The alternative hypothesis is, in each case, that at least one of the parameters considered is non-zero.

The F-test

- The F-ratio always has the mean square for error in the denominator.
- As before, it is a ratio of two variance estimates,
- and in each case, it can be seen as a signal-to-noise ratio: how large are the effects relative to the experimental error variance?

The test of $H_{AC} : (AC)_{ij} = 0$

- If the interaction term is significant:

The effect of A is different at each level of C or the effect of C is different at each level of A

- Interpretation of main effects is not very sensible.

Example

	B = Low	B = High
A = Low	10	20
A = High	20	10

Example of factorial studies: interactions

- The effect of advertising medium (Email list, instagram and television ads) and discount level (10% and 20%) on product sales.
- Suppose we find that:
 1. For email ads, increasing the discount from 10% to 20% leads to a small increase in sales.
 2. For instagram ads, the increase in discount leads to a large increase in sales.
 3. But for TV ads, sales are high regardless of the discount level.
- Then effect of discount (whether that is to increase, keep the same, or decrease) sales depends is not the same across the ad medium.

Example of factorial studies: interactions

- The effect of fertiliser type (F1,F2) and irrigation level (I1,I2) on wheat yield.
- Suppose we find that:
 1. With F1, changing irrigation from I1 to I2 gives a moderate yield increase.
 2. But with F2, the same change in irrigation leads to a very large yield increase.
- Then the effect of irrigation level on yield depends on the fertiliser used.

Advertising example

Sales per Ad Medium and Discount Level
(Wide Format)

Ad_Medium	10%	20%
Email	104, 105, 108	107, 110, 109
Instagram	104, 104, 107	121, 120, 119
TV	119, 115, 117	116, 113, 108

- How many treatments do we have?
- How many replicates do we have?

Calculating and interpreting effects

- $i = 1$ (Email), 2 (Insta), 3 (TV)
- $j = 1$ (10%), 2 (20%)

Treatment means

$$\bar{Y}_{11} = \frac{104+105+108}{3} = 105.66\dots$$

$$\bar{Y}_{12} = \frac{107+110+109}{3} = 108.66\dots$$

$$\bar{Y}_{21} = \frac{104+104+107}{3} = 105$$

$$\bar{Y}_{22} = \frac{121+120+119}{3} = 120$$

$$\bar{Y}_{31} = \frac{119+115+117}{3} = 117$$

$$\bar{Y}_{32} = \frac{116+113+108}{3} = 112.33\dots$$

Calculating and interpreting effects

Overall mean

$$\bar{Y}_{...} = \frac{105.66+108.66+105+120+117+112.33..}{6} = 111.44..$$

Main effects of Ad medium (A)

A_1 (Email)

What is the mean sales for Email ads? So the average response for the level Email.

$$= \frac{105.66..+108.66..}{2} = 107.166..$$

Subtract overall mean

$$A_1 = 107.166.. - 111.44... \approx -4.278$$

Calculating and interpreting effects

Main effects of Ad medium (A).

A_2 (Insta)

What is the mean sales for Instagram ads?

$$\hat{\mu}_1 = \frac{105+120}{2} = 112.50$$

Subtract overall mean

$$A_1 = 112.50 - 111.44... \approx 1.056$$

Calculating and interpreting effects

Main effects of Ad medium (A).

A_3 (TV)

What is the mean sales for TV ads?

$$\hat{\mu}_3 = \frac{117 + 112.33..}{2} = 114.66..$$

Subtract overall mean

$$A_1 = 114.66.. - 111.44.. \approx 3.222$$

Intepretation

- $A_1 = -4.278$: Mean sales using Email ads are lower than the overall mean, on average.
- $A_2 = 1.056$: Mean sales using Instagram ads are very close to average — just slightly above, on average.
- $A_3 = 3.22$: TV ads yield roughly 3 units more than the grand mean, on average.

Repeat for discount level

Main effects of Discount level (D).

$$D_1 = \hat{\mu}_1 - \hat{\mu} = \frac{105.66.. + 105 + 117}{3} - 111.44.. \approx -2.22$$

$$D_2 = \hat{\mu}_2 - \hat{\mu} = \frac{108.66.. + 120 + 112.33..}{3} - 111.44.. \approx 2.22$$

Repeat for discount level

- $D_1 = -2.22$: On average, sales when the discount level of 10% was applied, were less than the overall mean sales.
- $D_1 = 2.22$: On average, sales when the discount level of 20% was applied, were more than the overall mean sales.

Interaction effect

$$(AD)_{ij} = \bar{Y}_{ij.} - (\mu + A_i + D_j)$$

- Treatment 11 (Email and 10%)

$$(AD)_{11} = \bar{Y}_{11.} - (\mu + A_1 + D_1) = 105.666 - (111.44.. - 4.72.. - 2.22..) \approx 0.722$$

- Treatment 32 (TV and 20%)

$$(AD)_{32} = \bar{Y}_{32.} - (\mu + A_3 + D_2) = 112.33.. - (111.44.. + 3.22.. + 2.22..) \approx -4.556$$

Interpretation

- An interaction effect shows how much a particular treatment combination (e.g. Instagram + 20% discount) deviates from what you would expect based on adding the main effects alone.
- If there was no interaction, then the effect of one factor (like discount level) would be the same regardless of the level of the other factor (ad medium).
- So a non-zero interaction effect means the factors don't operate independently.

Example

- $(AD)_{11} = 0.722$
- When using email ads with 10%, the mean sales is more than expected when considering the main effects alone.
- $(AD)_{32} = -4.556$
- When using TV ads with 20% discount, the mean sales is slightly less than expected if the factors didn't interact.
- These are a bit abstract and difficult to understand, to understand the interaction we need to look at a plot.

But first, how to fit this model in R

```
1 fact_model <- aov(Sales ~ Ad_Medium + Discount_Level
2                   + Ad_Medium:Discount_Level, data = sales_data)
3
4 # OR
5
6 fact_model <- aov(Sales ~ Ad_Medium*Discount_Level, data = sales_data)
```

Extract estimates

```
1 means <- model.tables(fact_model, type ="means")
```

Output

```
> model.tables(full_model, type = means)
```

Tables of means

Grand mean

111.4444

Ad_Medium

Ad_Medium

Email	Instagram	TV
107.17	112.50	114.67

Discount_Level

Discount_Level

10%	20%
109.22	113.67

Ad_Medium:Discount_Level

	Discount_Level	
Ad_Medium	10%	20%
Email	105.67	108.67
Instagram	105.00	120.00
TV	117.00	112.33

Extract estimates

```
1 effects <- model.tables(fact_model, type ="effects")
```

Output

Tables of effects

Ad_Medium
Ad_Medium

Email	Instagram	TV
-4.278	1.056	3.222

Discount_Level
Discount_Level

10%	20%
-2.2222	2.2222

Ad_Medium:Discount_Level
Discount_Level

Ad_Medium	10%	20%
Email	0.722	-0.722
Instagram	-5.278	5.278
TV	4.556	-4.556

Sums of squares

Each Sum of Squares (SS) of the decomposition measures how much variation is explained by one component of the model.

- SST: Total variation in all observations around the grand mean.
- SSA: Variation due to differences between levels of Factor A (Ad Medium), ignoring Factor D.
- SSD: Variation due to differences between levels of Factor D (Discount), ignoring Factor A.
- SSAD: Variation due to the interaction effects.
- SSE: Variation within each treatment group — the random, unexplained variation.

ANOVA Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Ad_Medium	2	178.78	89.39	17.30	0.000292	***
Discount_Level	1	88.89	88.89	17.20	0.001352	**
Ad_Medium:Discount_Level	2	294.78	147.39	28.53	2.75e-05	***
Residuals	12	62.00	5.17			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Hypotheses

Interaction:

- H_0 : All interaction effects are equal to zero, the effect of Advertising Medium does not depend on the level of Discount.
- H_A : At least one interaction effect is not zero, the effect of Advertising Medium does depend on the level of Discount (there is an interaction).

Hypotheses

Main effects A:

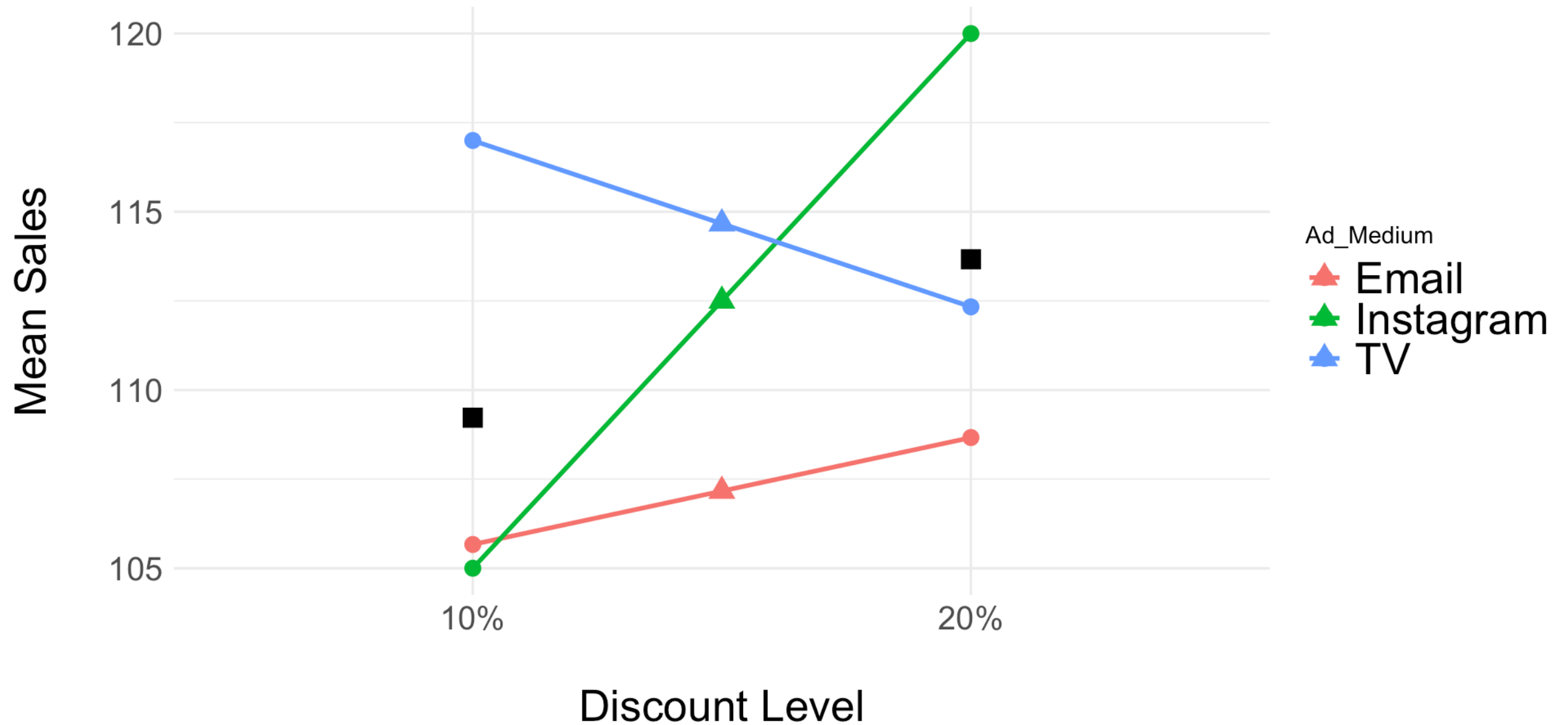
- H_0 : All main effects of A are zero, there is no difference in average sales between the levels of advertising medium.
- H_A : At least one main effect is not zero, one (or more) of the advertising mediums resulted in a different mean sales.

Hypotheses

Main effects D:

- H_0 : All main effects of D are zero, there is no difference in average sales between the levels of discount level.
- H_A : At least one main effect is not zero, one (or more) of the discount percentages resulted in a different mean sales.

Interaction plot



In words

- The effect of Discount (i.e. difference between 10% and 20%) differs depending on the advertising medium used.
- For example, the sales decreases between 10% and 20% for TV ads, but for Instagram and Email ads it increases (and more so for Instagram ads).
- Or vice versa, the effect Advertising medium (i.e. difference between the levels) differs depending on the discount level.